

Epithelium

Specializations (polarity of epithelium)

Apical surface :-

faces the lumen or the external environment

" *Microvilli, cilia, stereocilia* "

Lateral surface:-

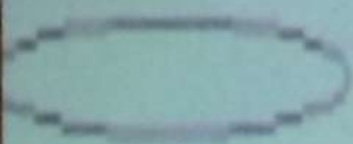
faces the sides of adjacent cells

" *Tight junctions (zona occludens), adherens junction (zona adherens), desmosomes (macula adherens), gap junctions* "

Basal surface

attaches to the basement membrane

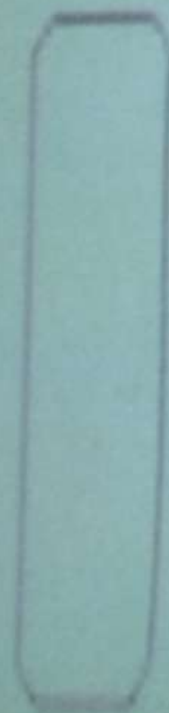
" *Basement membrane, hemidesmosomes* "



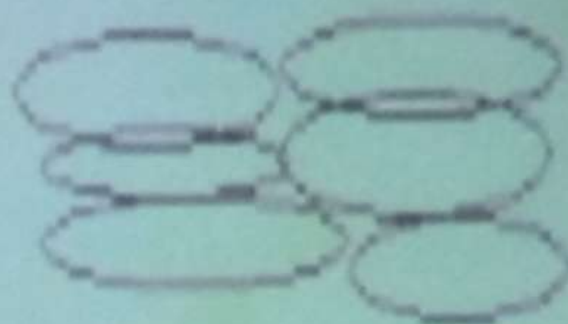
Squamous



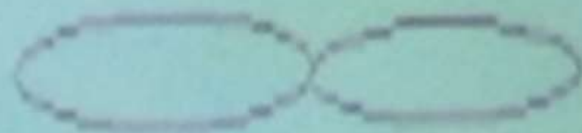
Cuboidal



Columnar



Stratified

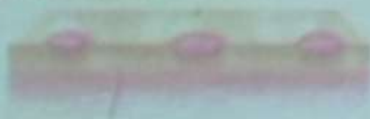


Simple

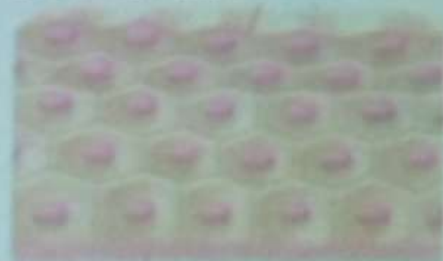
CLASSIFICATION OF EPITHELIUM

On the basis of :

NUMBER OF LAYERS



SIMPLE (made of single cell layer)



STRATIFIED (made of many cell layer)

CELL SHAPE



Squamous



Cuboidal



Columnar



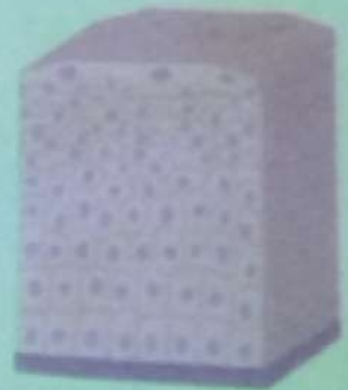
Simple Cuboidal



Simple Squamous



Simple Columnar

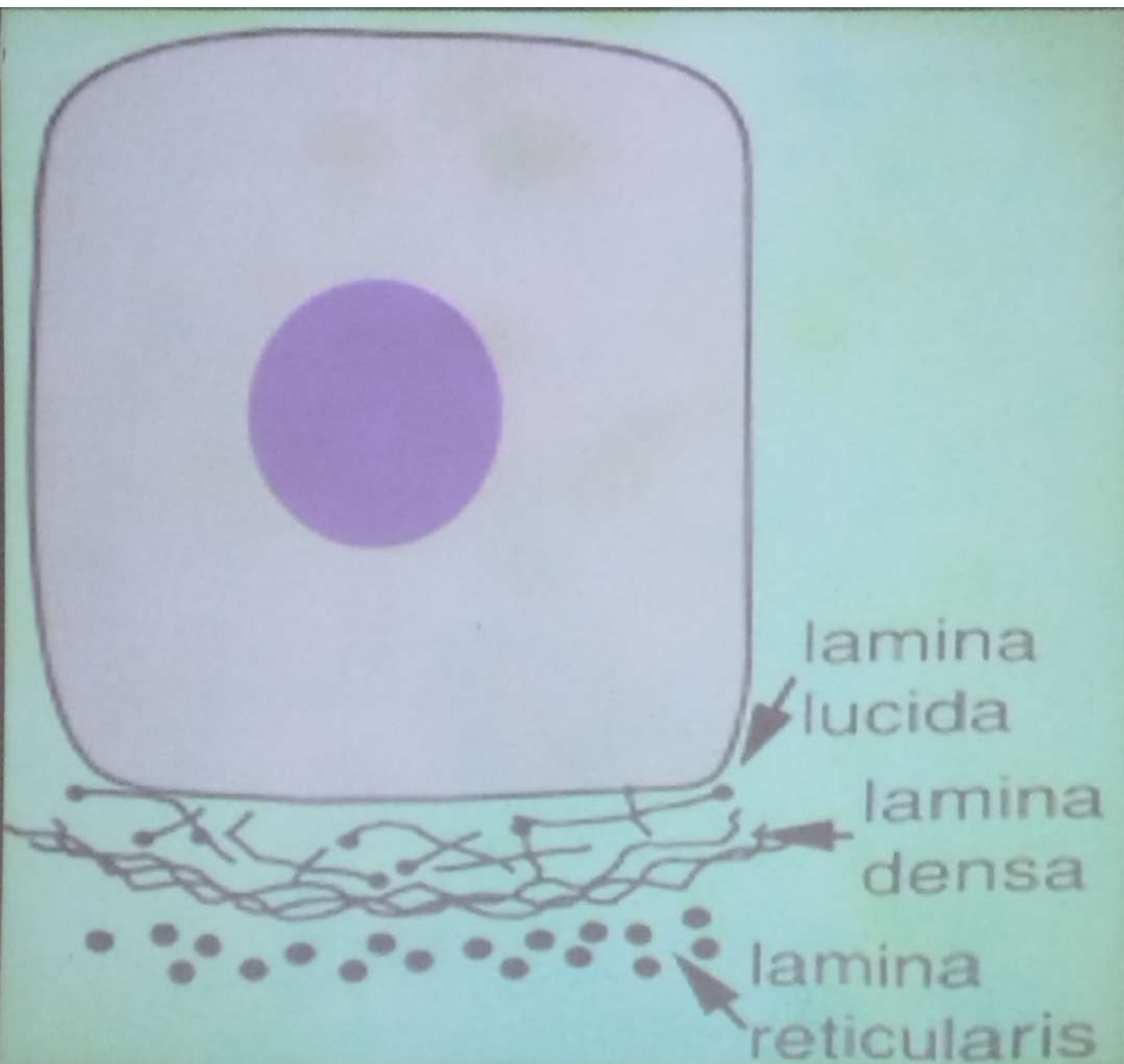


Stratified Squamous

Specilization of the basal surface of epithelium

A-The basement membrane:-

- is a thin, sheet-like type of extracellular matrix, that provides cell and tissue support
- The basement membrane sits between epithelium and the underlying connective tissue.
- As seen with the electron microscope, the basement membrane is composed of two layers, the basal lamina and the reticular lamina.
- The basal lamina layer can further be divided into two layers. The clear layer closer to the epithelium is called the lamina lucida, while the dense layer closer to the connective tissue is called the lamina densa.



B- Hemidesmosomes:-

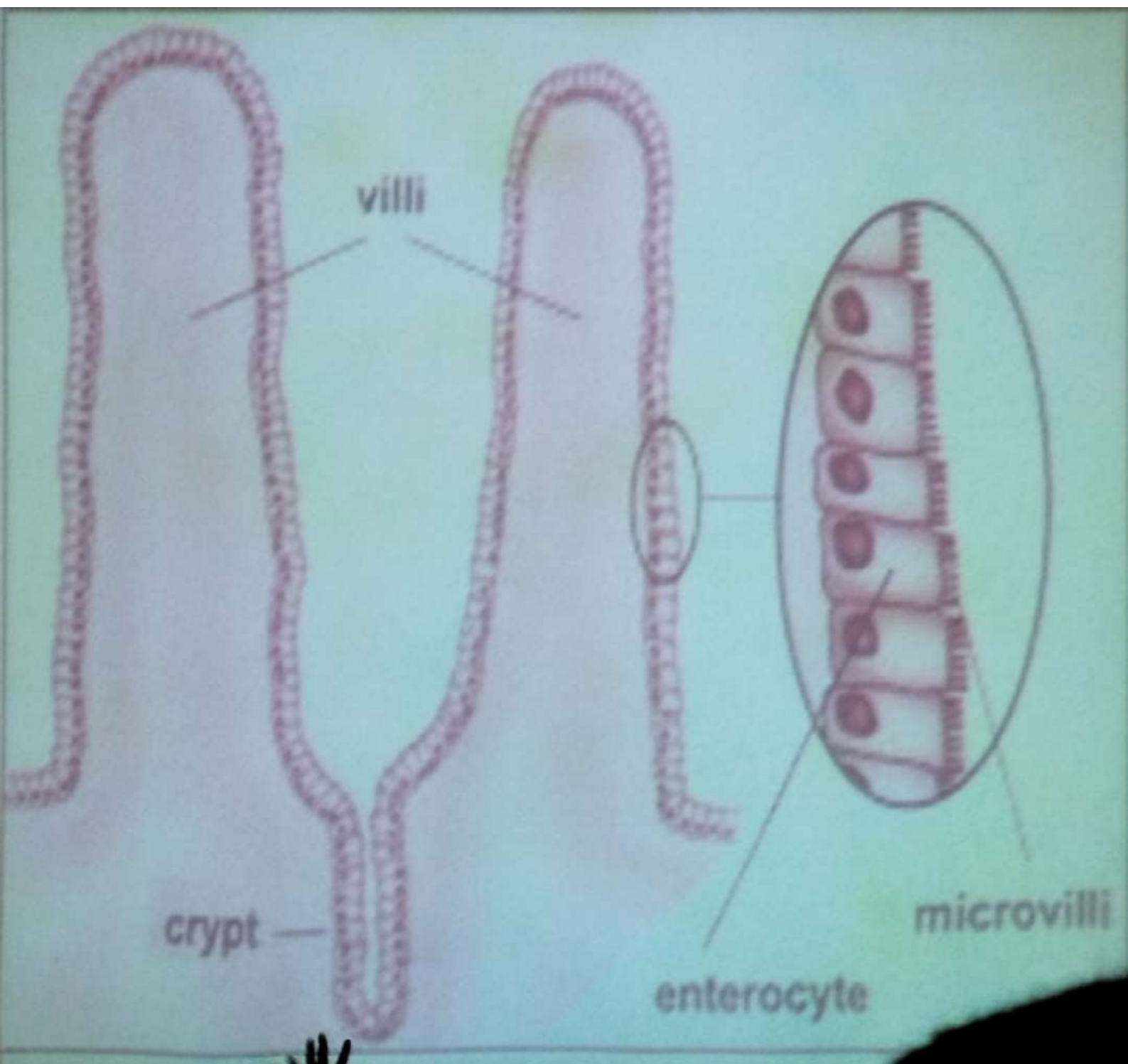
- ❑ These look similar to desmosomes, but are different functionally, and in their content.
- ❑ They connect the basal surface of epithelial cells via intermediate filaments to the underlying basal lamina.

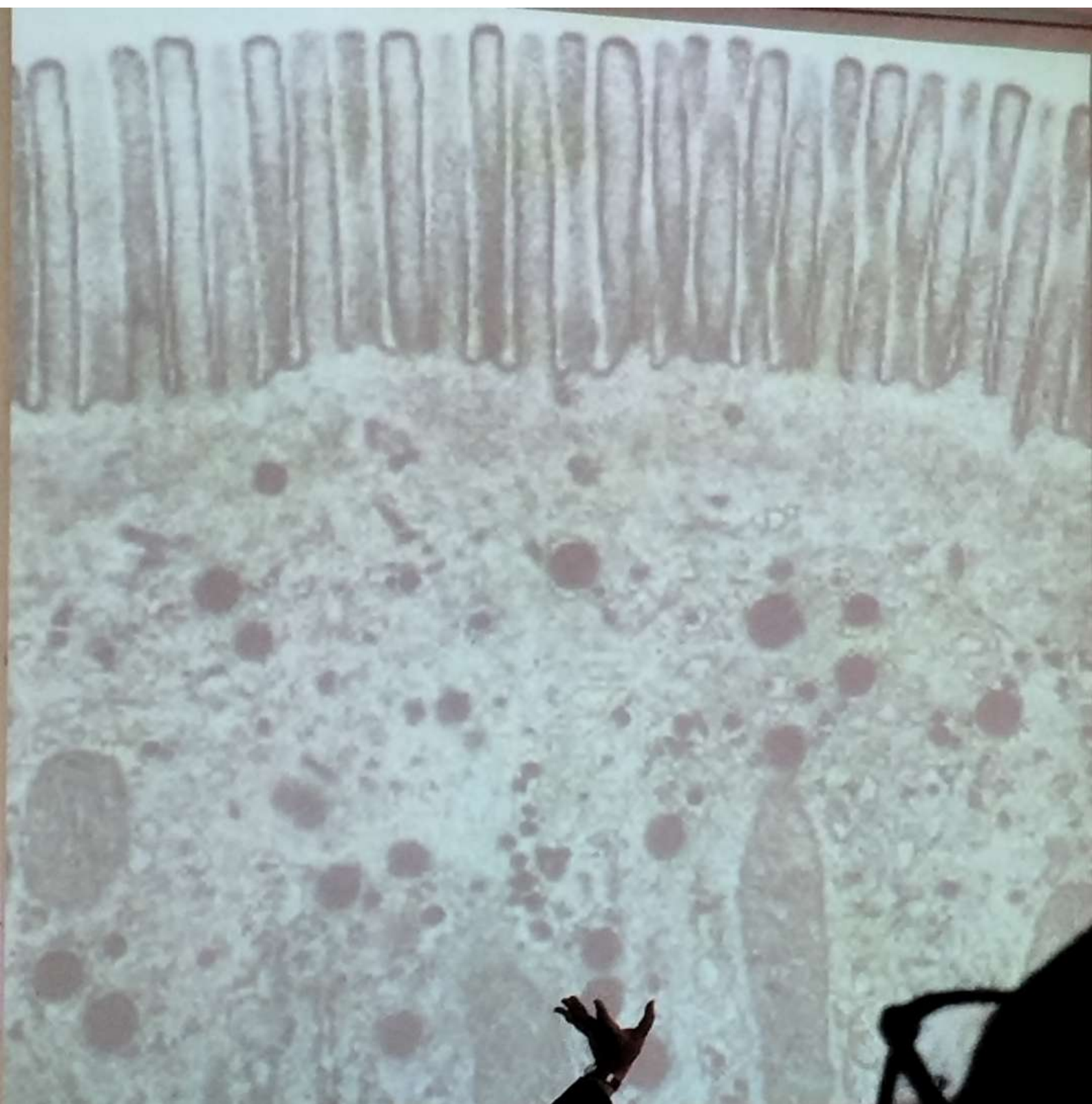


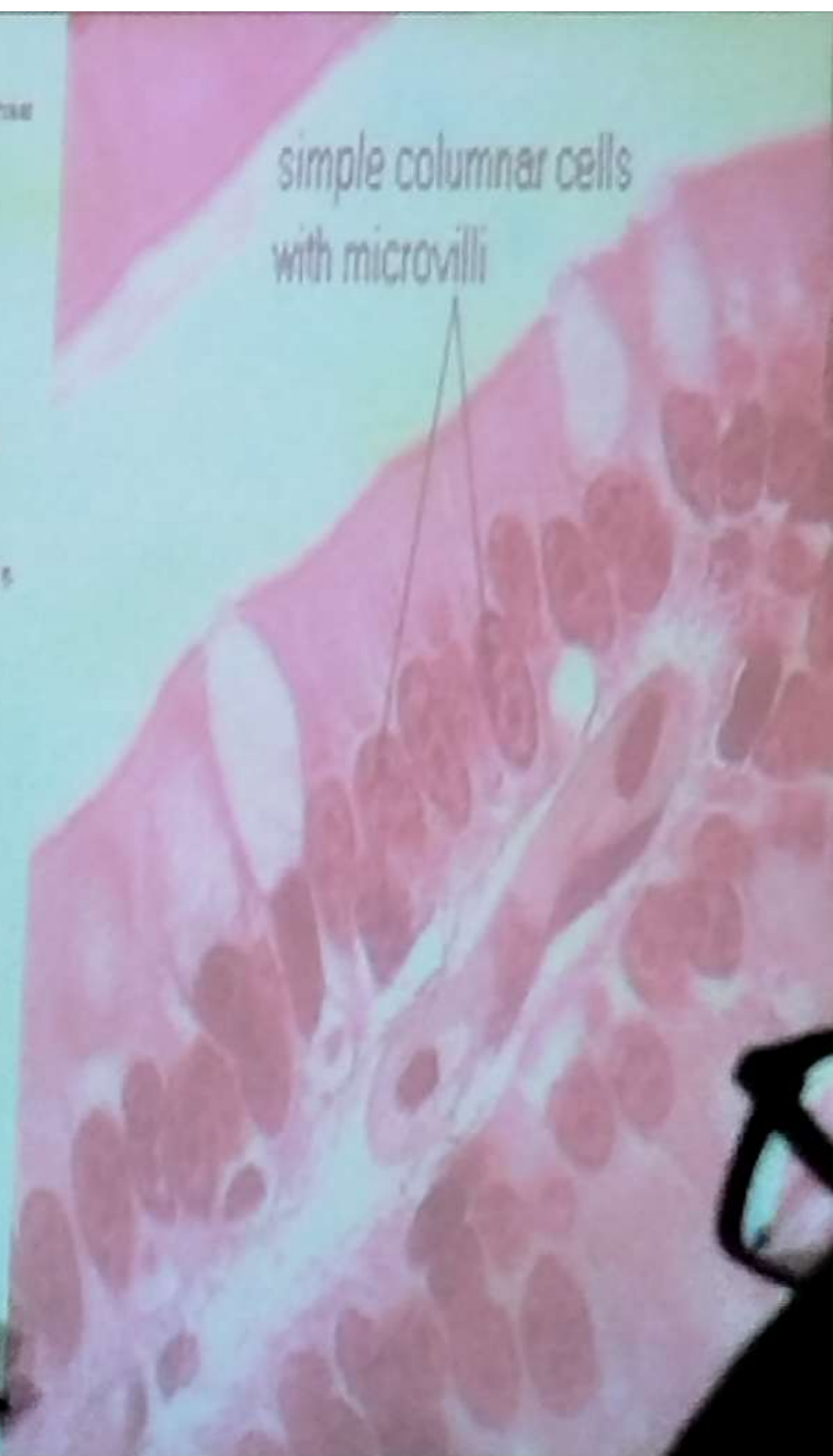
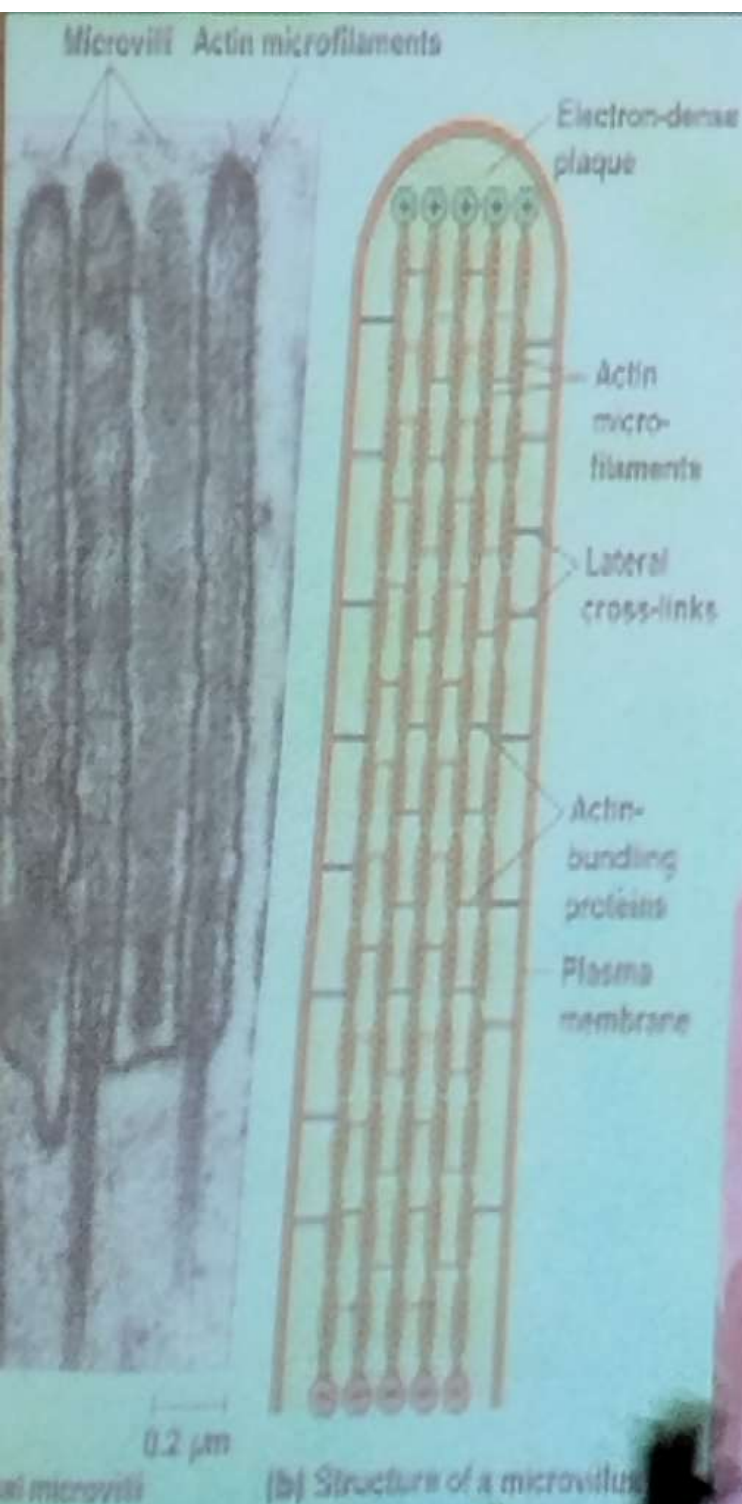
Specialization of the free portion of epithelium

a-Microvilli

- ❑ These are small finger like projections, about 1mm in length, and 90nm or so in diameter.
- ❑ The microvilli are shorter and narrower than cilia. They contain bundles of parallel actin filaments held together into a bundle by cross-linking proteins called villin and fimbrin.(
- ❑ Microvilli are present on the luminal surface of many epithelia, particularly those specialised for absorption. Microvilli increase the area of the free surface of the epithelium available for absorption.



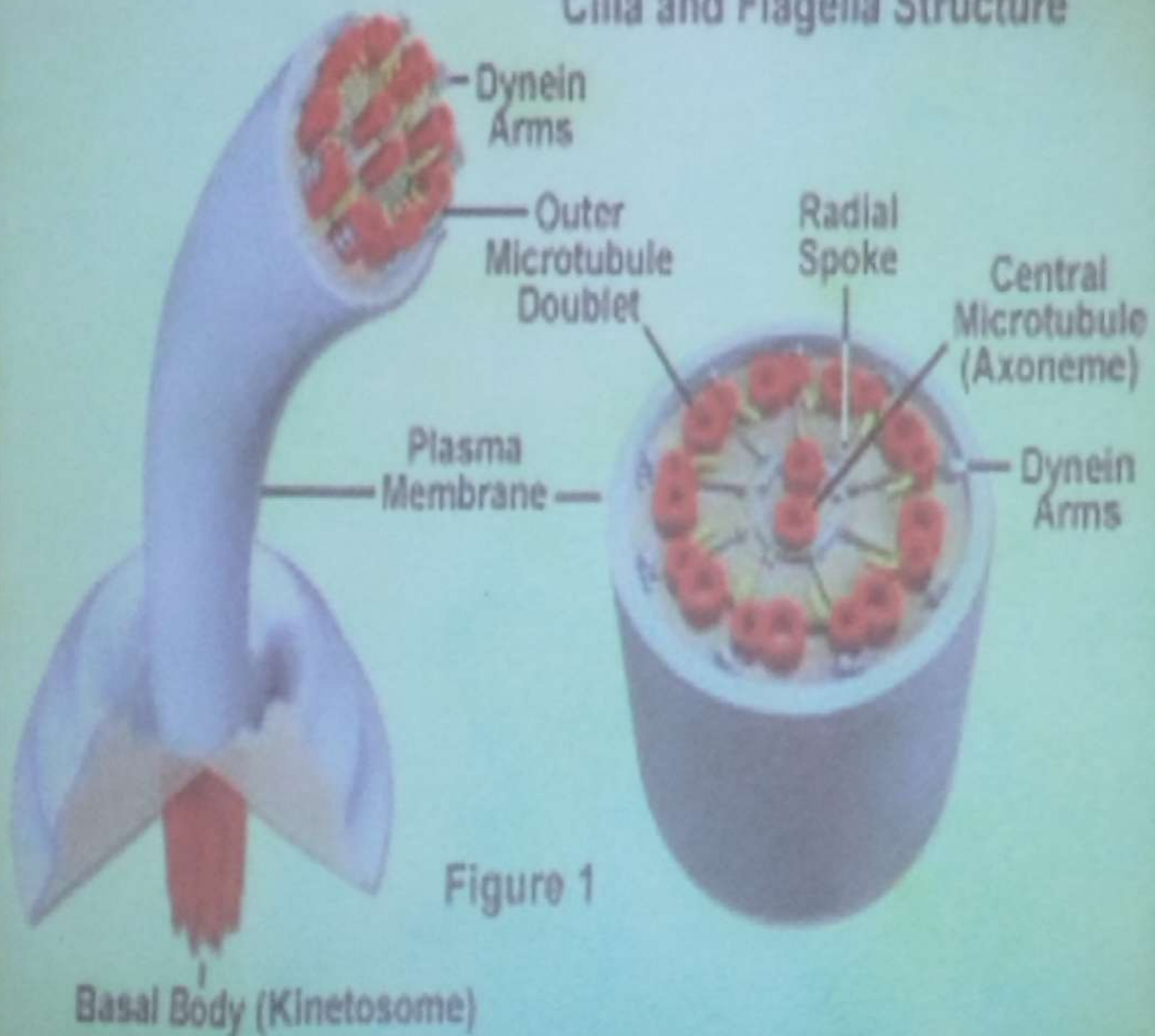




Cilia(kinocilia)

- Cilia are hair-like motile processes. They are about $10\mu\text{m}$ long and about $0.2\mu\text{m}$ in diameter, and are longer and thicker than microvilli. They are made of microtubules .
- There is a ring of doublet microtubules, and a central pair of singlet microtubules.
- At the base of the cilium is the basal body, into which the microtubules are anchored.

Cilia and Flagella Structure

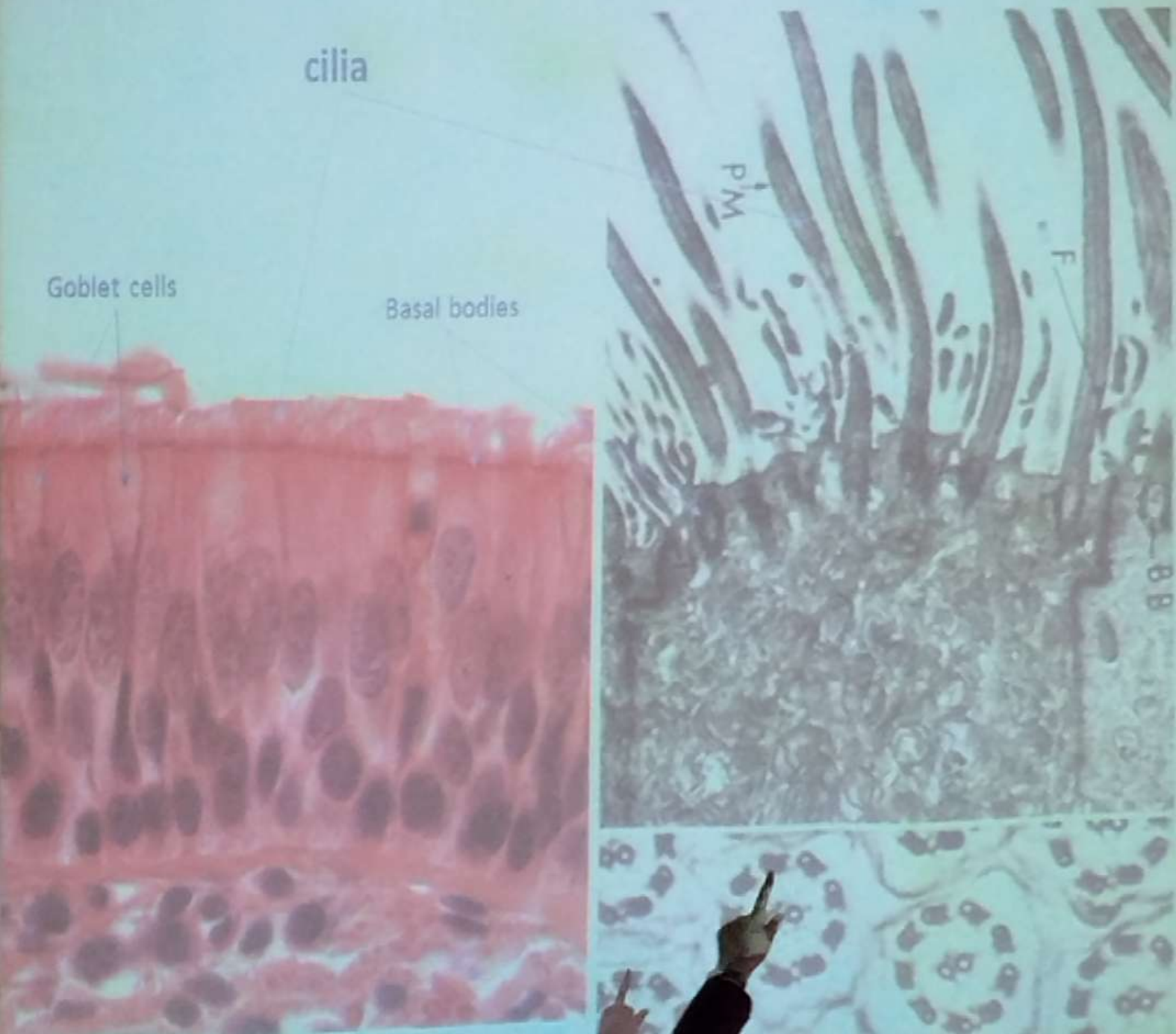


Cilia (Apical Cell Surface Specializations - 2)
core of microtubules in 9+2 arrangement (axoneme)

cilia

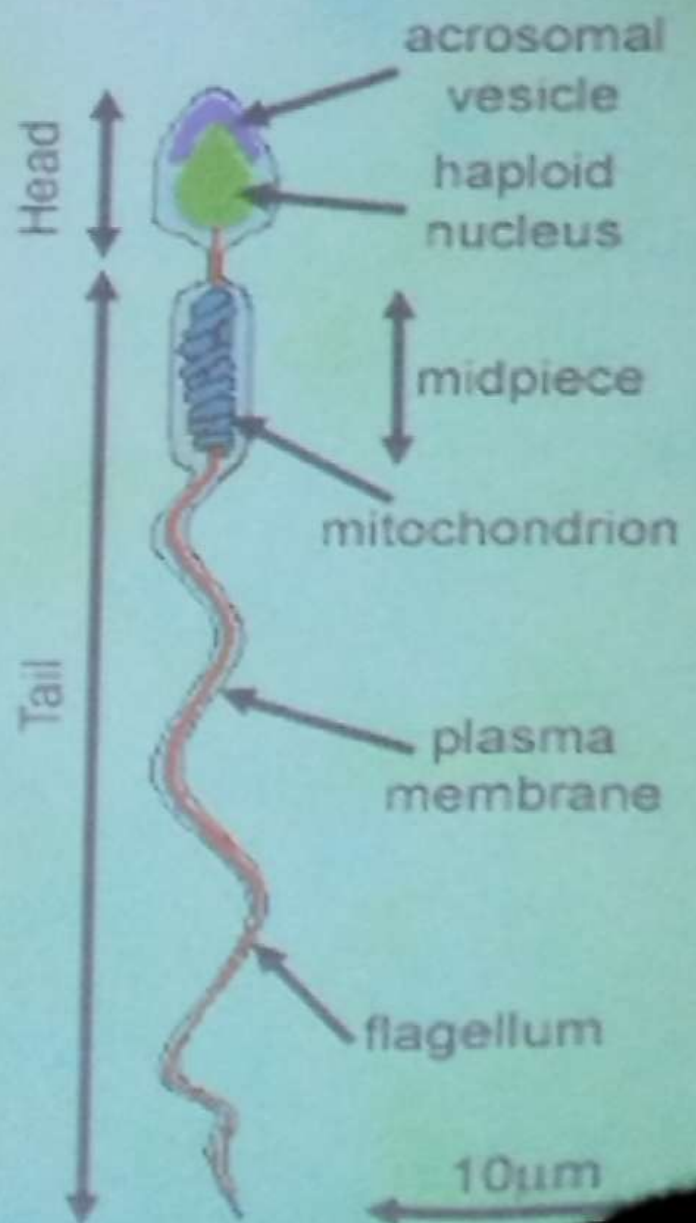
Goblet cells

Basal bodies



flagella

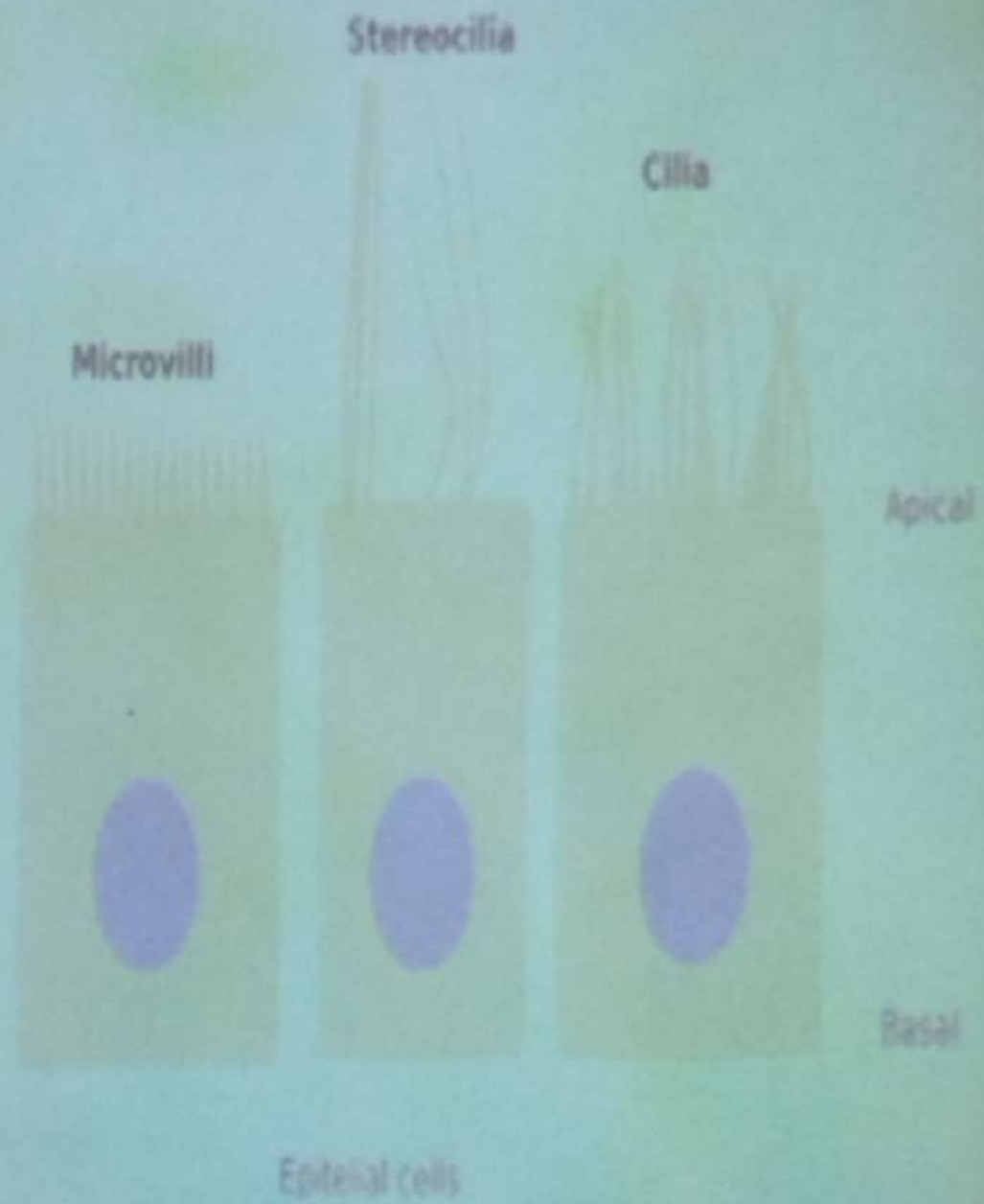
- Flagella are longer 100-200 μm .
- Flagella is present in sperm one per cell, while cilia are present in large number 300 cilia per cell.



Stereocilia

- They are not true cilia they are long nonmotile microvilli which project into the lumen from each cell of the epithelial lining the epididymis and ductus deferens. They lack filamentous cores and basal corpuscles.
- They regulate the specialized environment in which the spermatozoa can mature. They play a role in secretion and absorption of the surrounding medium.

	Microvilli	Cilia
Shape	Finger-like extensions	Hair-like projections
Site	Epithelial cells of the small intestine and kidney tubules	Epithelial cells of the respiratory tract and uterine tube
Motility	Non-motile	Motile
Function	Increase the surface area for absorption	Involved in movement (sperm clearance, and help moving the ovum)
Composition	Made up of microfilaments	Made up of microtubules
Number	Numerous	Fewer



Cross Sections



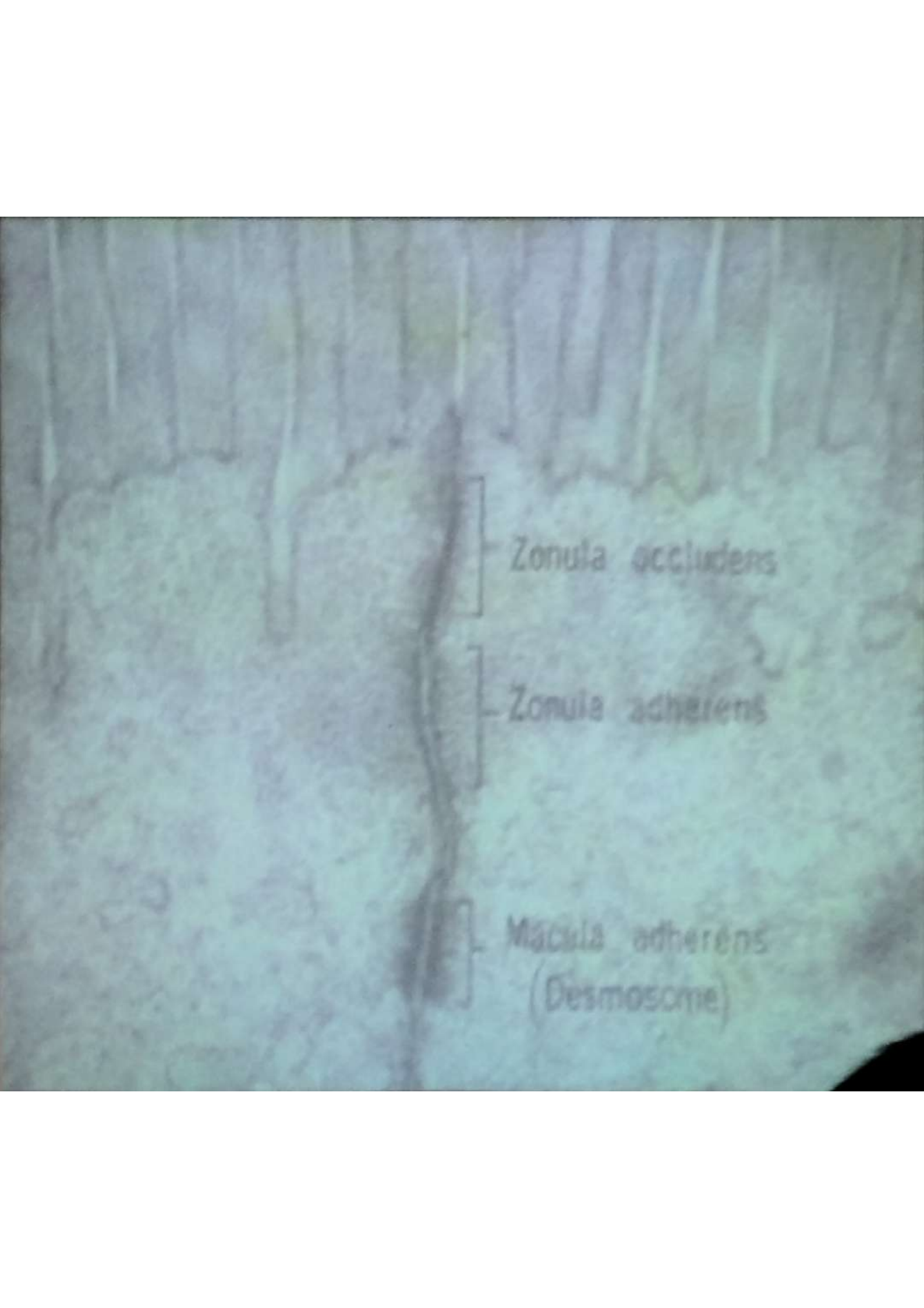
Villi of small intestine

Cilia

Microvilli

Specialization of the lateral surface

1. **Occluding junctions** (zonula occludens or tight junctions)
2. **Adhering junctions** (zonula adherens).
3. **Desmosomes** (macula adherens). There are also 'hemidesmosomes' that lie on the basal membrane, to help stick the cells to the underlying basal lamina.
4. **Gap junctions**. These are communicating junctions. (also known as nexus, septate junction).



This electron micrograph shows the interface between two adjacent cells. The image is characterized by a light, granular background with darker, more electron-dense structures. Three specific junctional complexes are highlighted with brackets and labels on the right side. The topmost label, 'Zonula occludens', points to a thin, dark horizontal line. Below it, the label 'Zonula adherens' points to a slightly thicker, more irregular dark line. The bottom label, 'Macula adherens (Desmosome)', points to a large, dark, and highly electron-dense area. The overall texture is typical of biological tissue at high magnification.

Zonula occludens

Zonula adherens

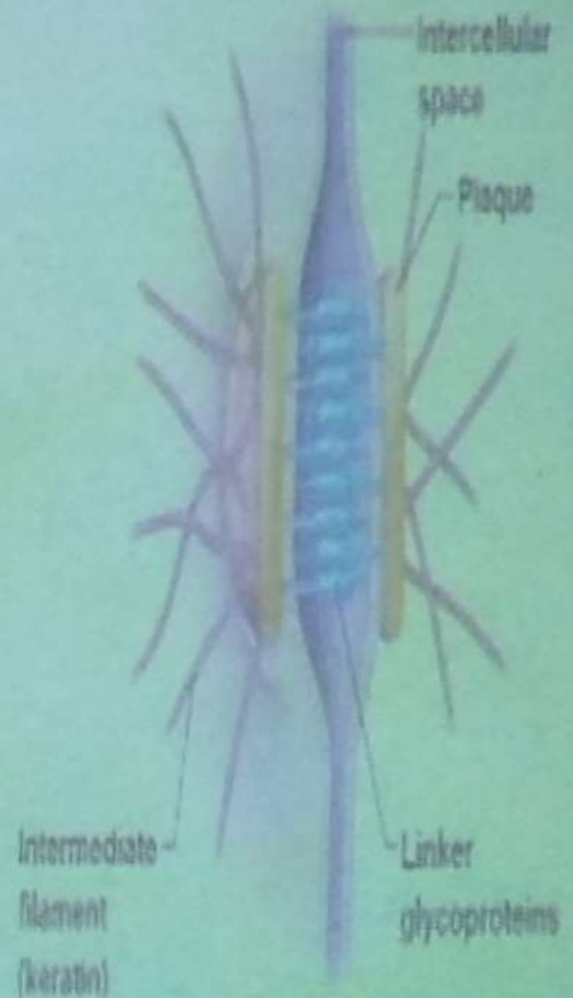
Macula adherens
(Desmosome)

• Desmosomes

Desmosomes connect two cells together. A desmosome is also known as a spot desmosome or macula adherens (macula = latin for spot), because it is circular or spot like in outline, and not belt- or band shaped like adherens junctions.

- There is dense material between the two cell membranes, which is made up of trans membrane linker glycoproteins, also the intermediate filaments running from the desmosome into the cytoplasm.
- It is made up of a dense cytoplasmic plaque, to which the intermediate filaments attach.

Desmosome



(b) Desmosomes: Anchoring junctions bind adjacent cells together and help form an internal tension-reducing network of fibers.

Classification of Epithelial tissue

Epithelial Tissue

Simple Epithelium

Squamous Epithelium

Cuboidal Epithelium

Columnar Epithelium

Ciliated Epithelium

Pseudo-stratified Epithelium

Compound Epithelium

Stratified Compound

Transitional Compound

Glandular Epithelium



CEBT

THE CENTRAL EPITHELIAL TISSUE BOARD

I. Simple Epithelium :-

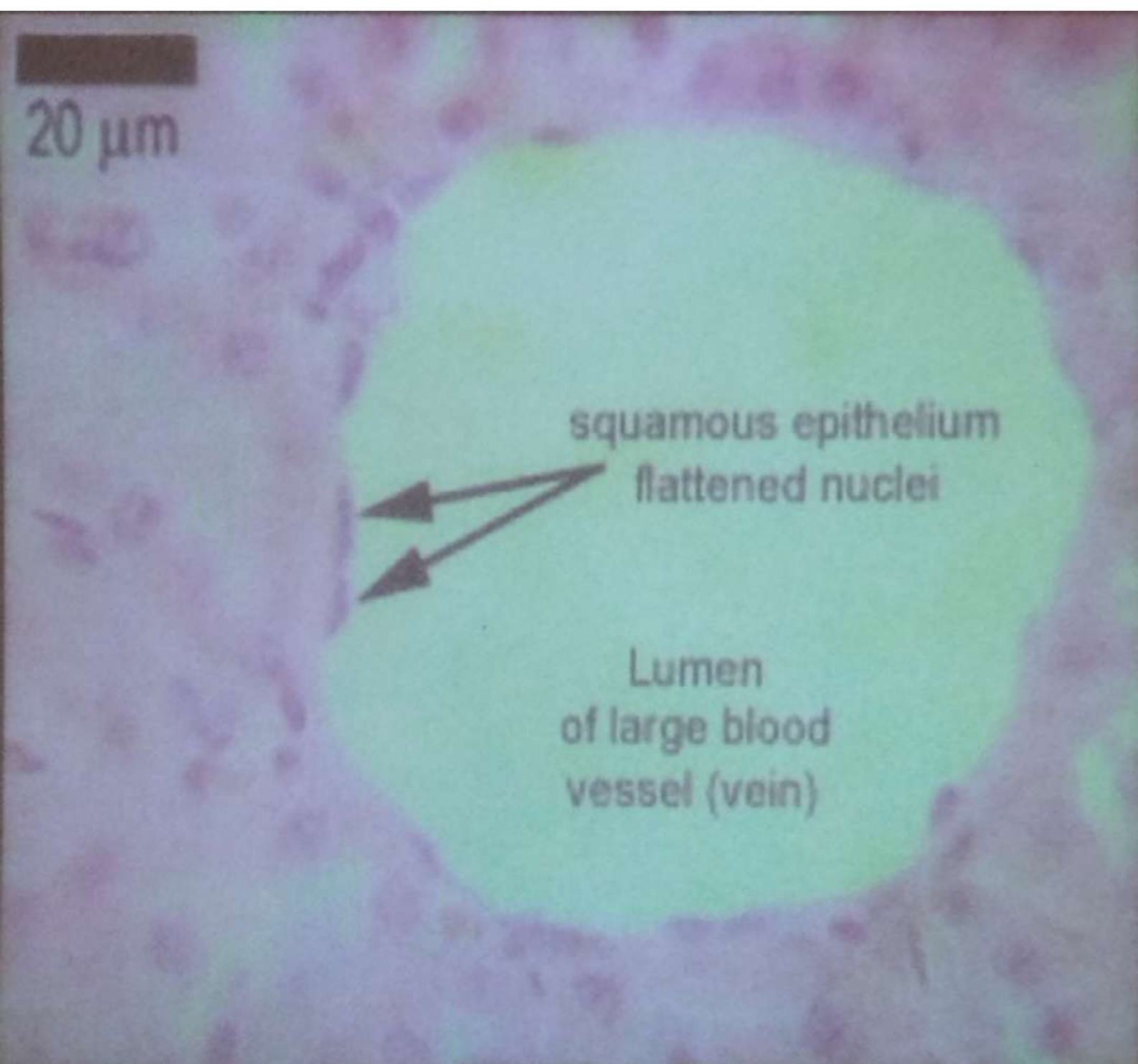
A-Simple squamous epithelium

- consists of a single layer of flattened cells. The thinness of these cells facilitates the transfer of materials (*e.g.*, gases, fluids or nutrients) across the epithelium.

20 μm

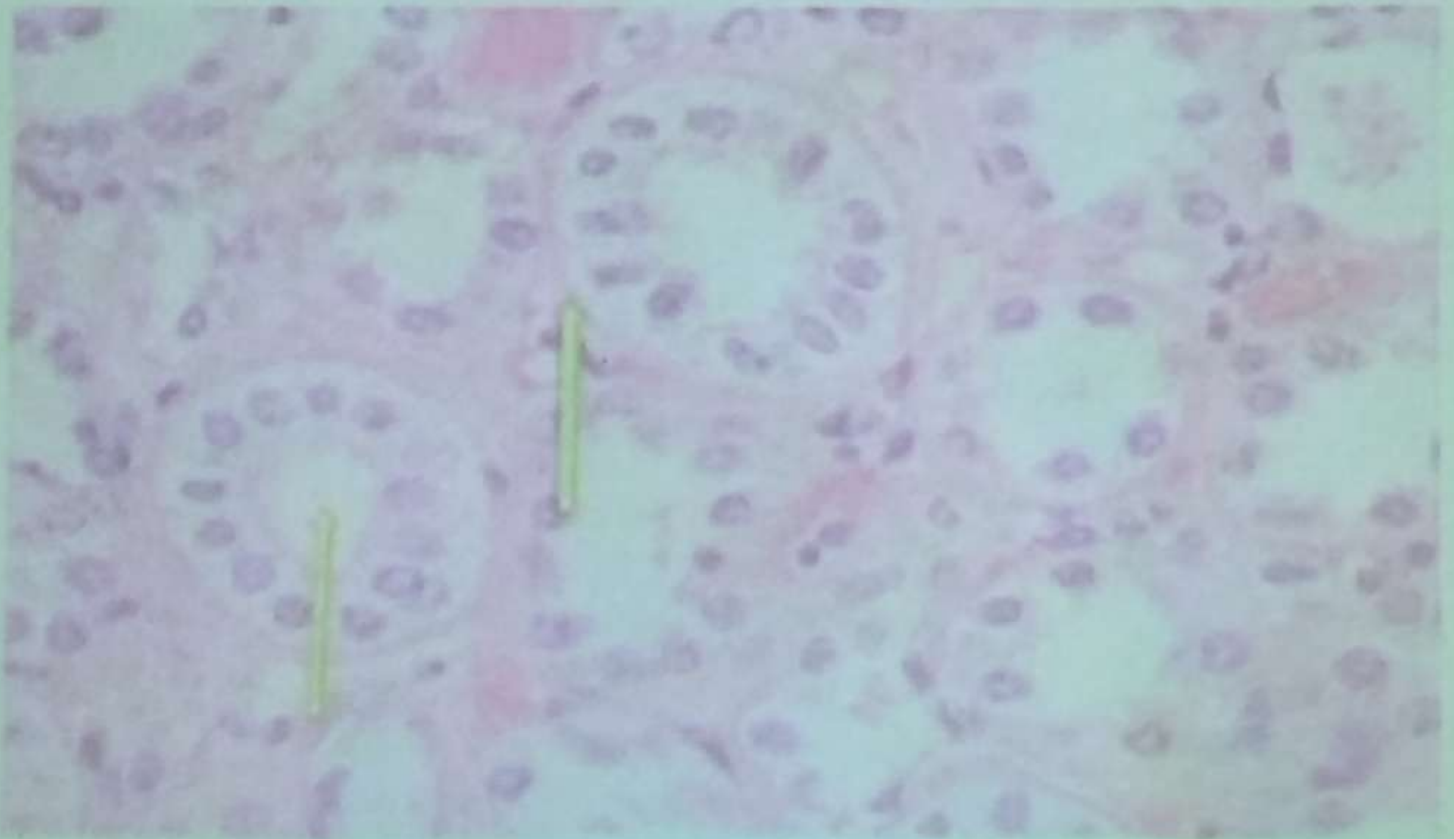
squamous epithelium
flattened nuclei

Lumen
of large blood
vessel (vein)

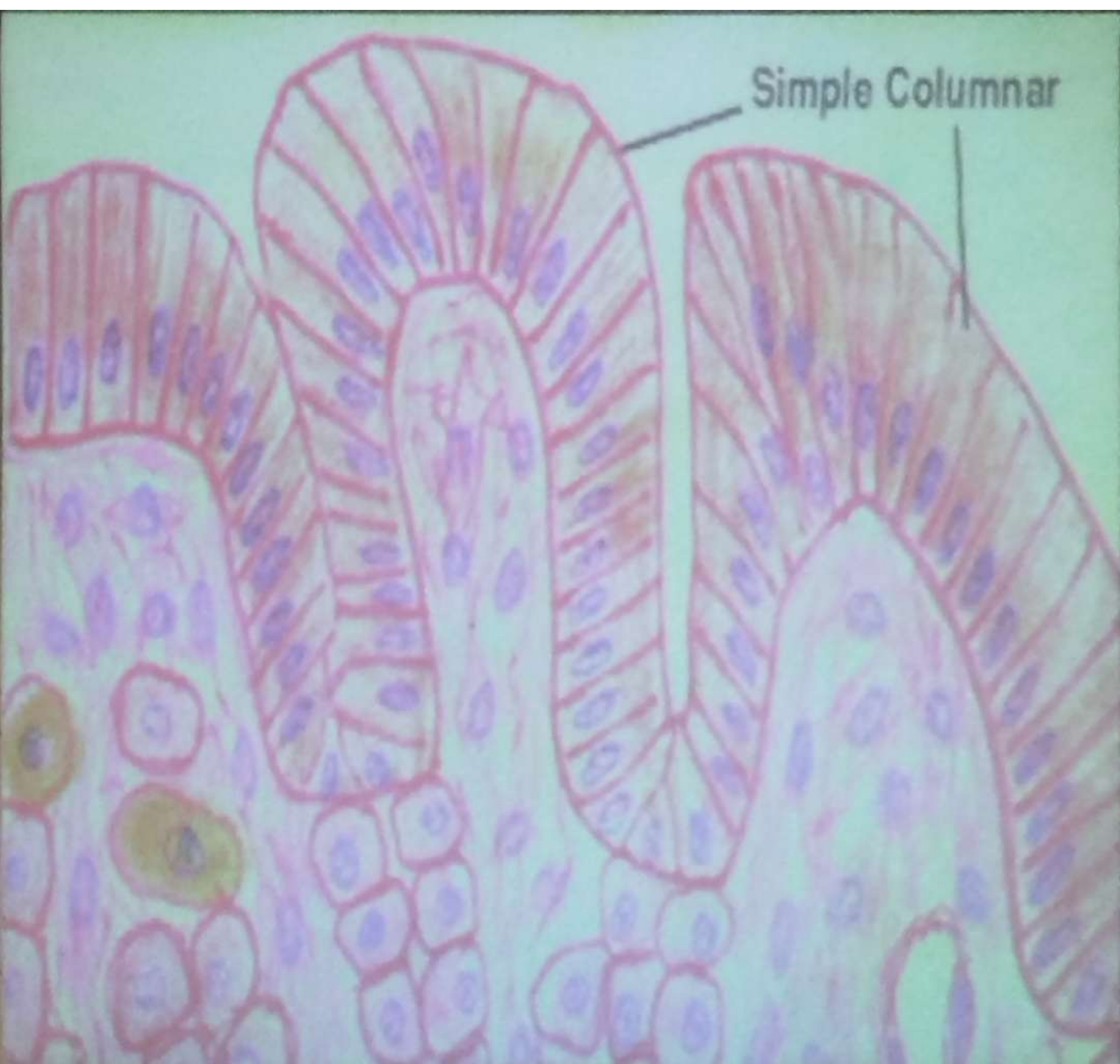


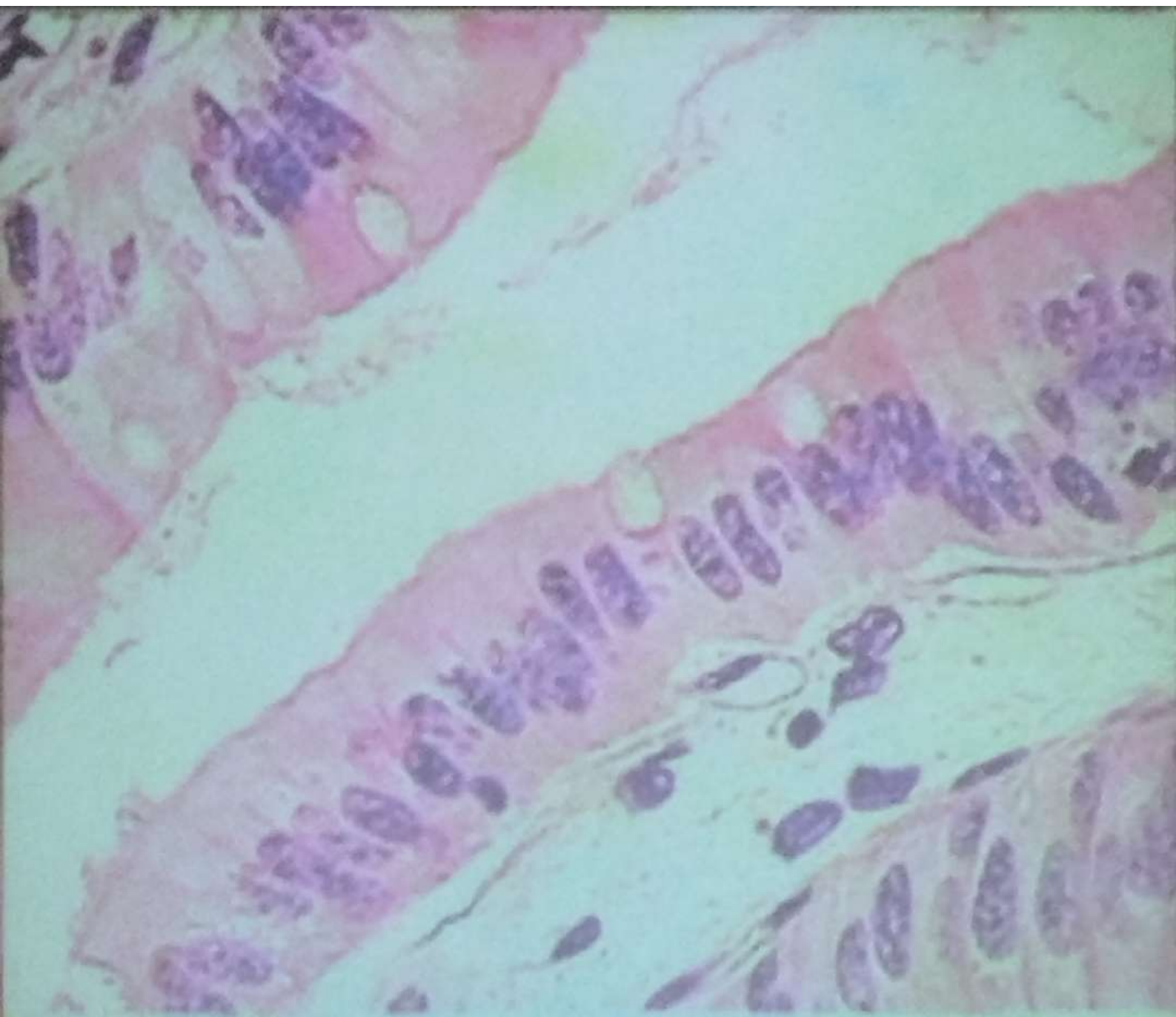
B-Simple cuboidal epithelium

Simple Cuboidal Epithelium



Identify the structures at the arrows.





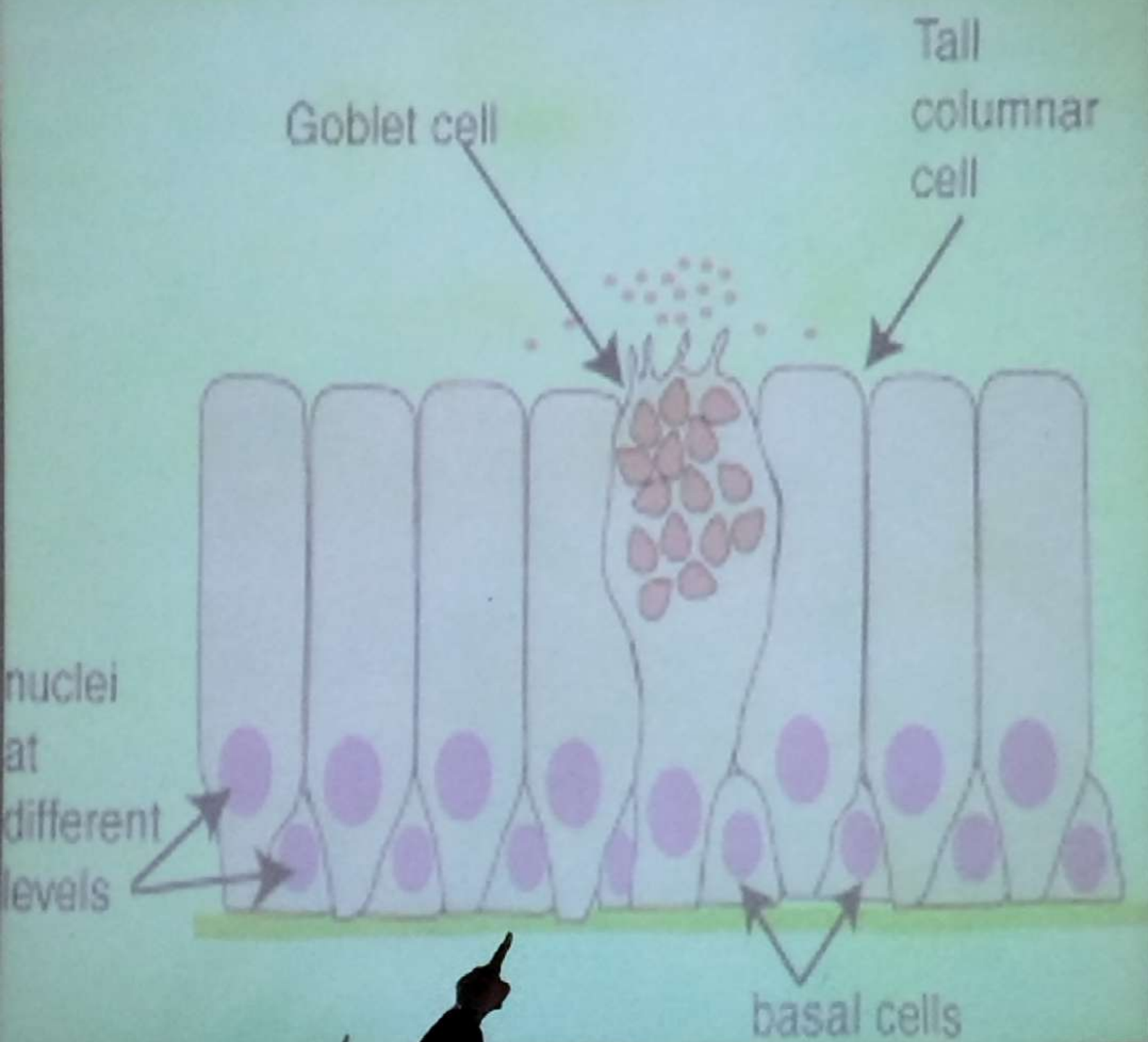
Simple Columnar Epithelium - Small Intestine

D-Pseudostratified Columnar Epithelium

- one layer of cells, but the nuclei are at different heights, so it looks as though it is more than one layer. These epithelia usually have goblet cells present.

Occurrence: •

- a. *The non ciliated pseudostratified columnar epithelium* occurs •
in the large duct of the salivary gland, vas deferens, and in the male urethra .
- b. *Ciliated pseudostratified columnar epithelium* is found in the •
greater part of respiratory passages and in the epididymis.

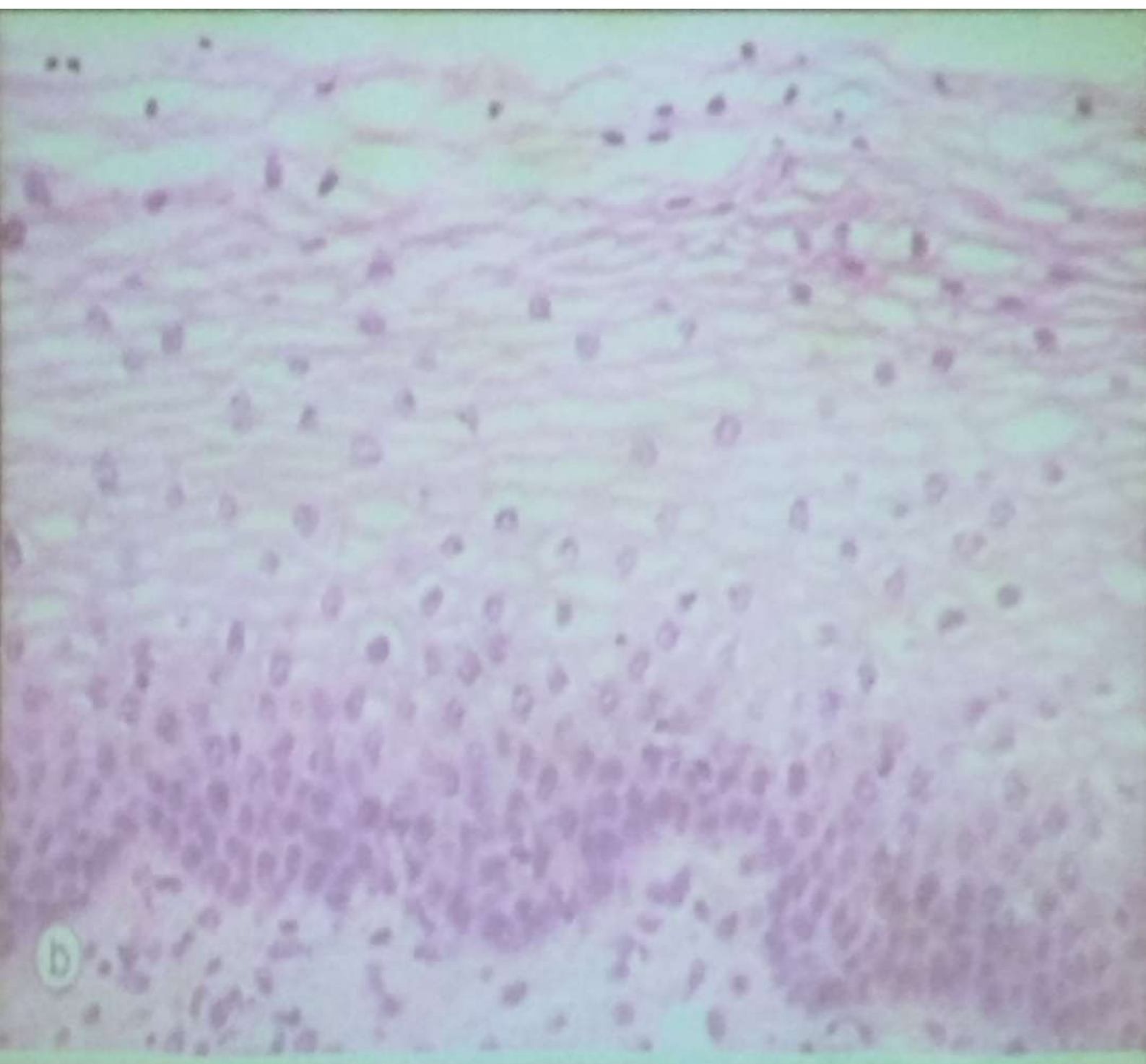


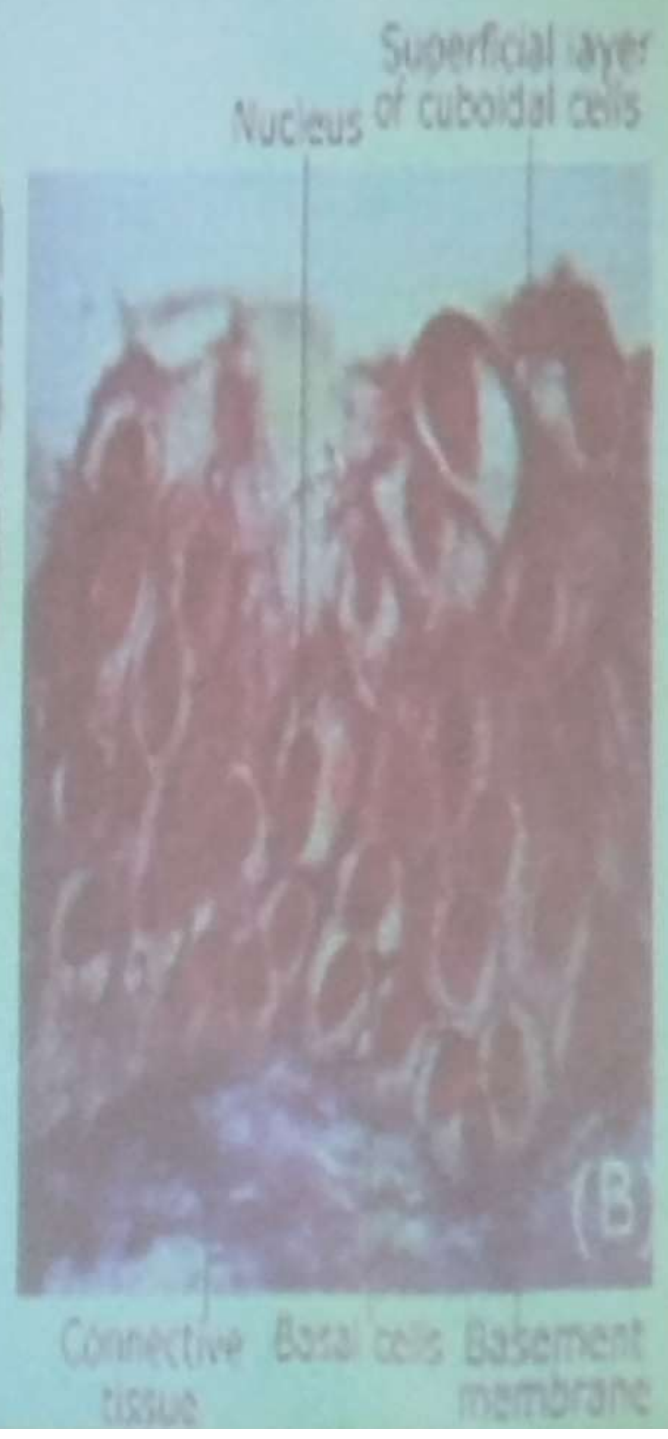
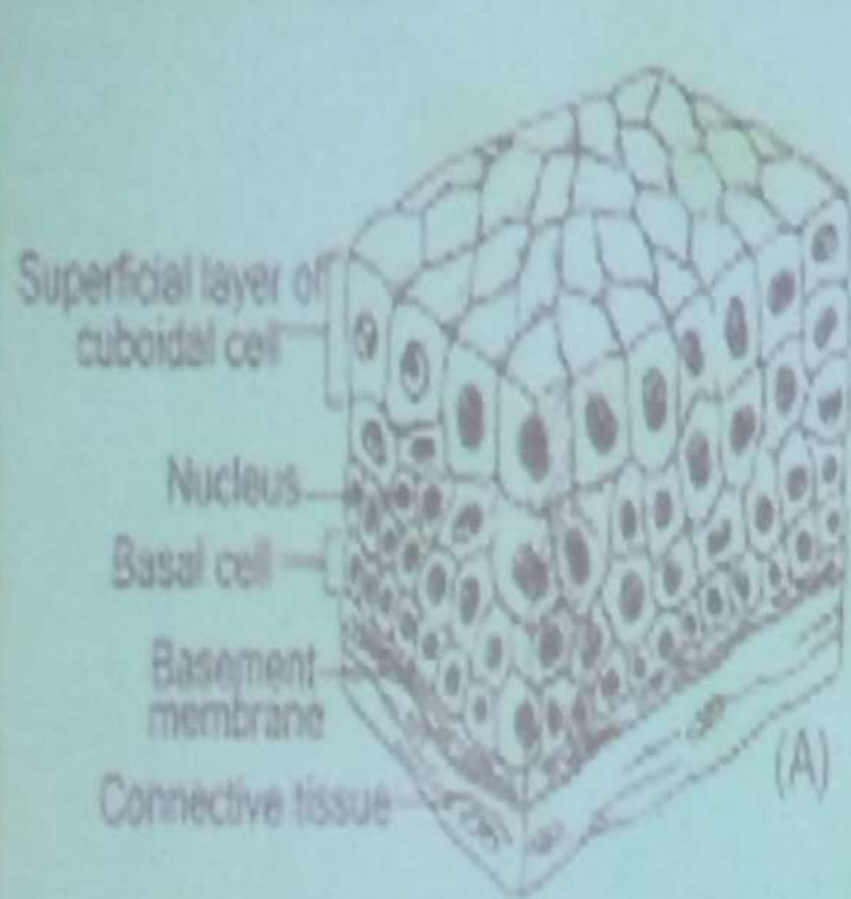
II. Stratified Epithelium :

A. Stratified squamous Epithelium : *keratinized or non keratinized*

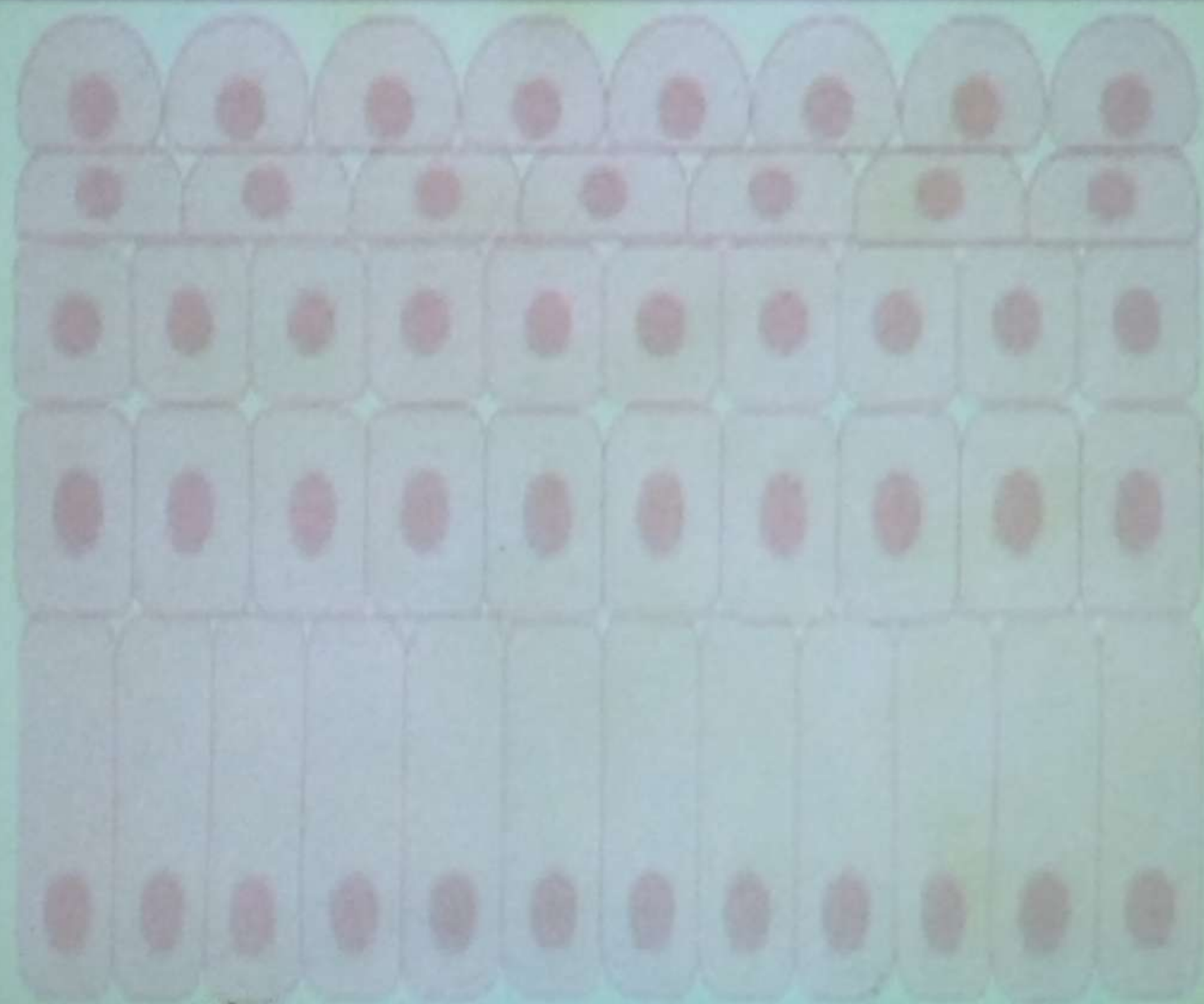
1. Stratified squamous epithelium: (non keratinized): It consists of three layers:

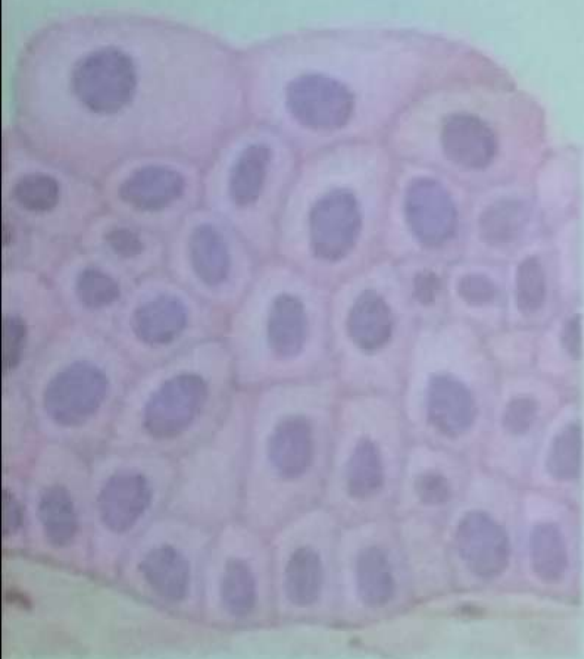
- a. The layer lying on the basement membrane is composed of columnar cells with an oval nucleus.
- b. In the middle layers, the cells are polyhedral with a spherical nucleus.
- c. The superficial layers are flattened cells and contain a flat nucleus.
- **Occurrence:**
- In the esophagus, vagina, cornea, and the mucocutaneous junctions



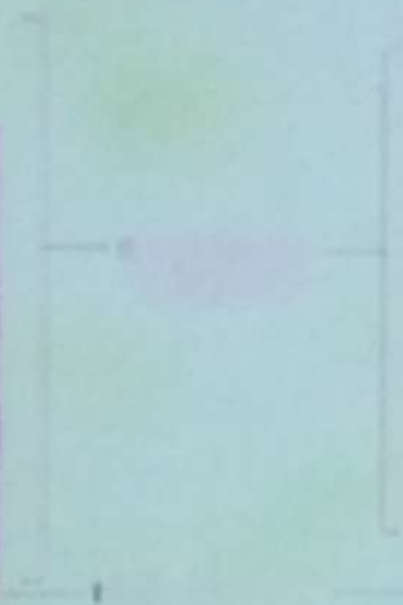


Transitional Epithelium





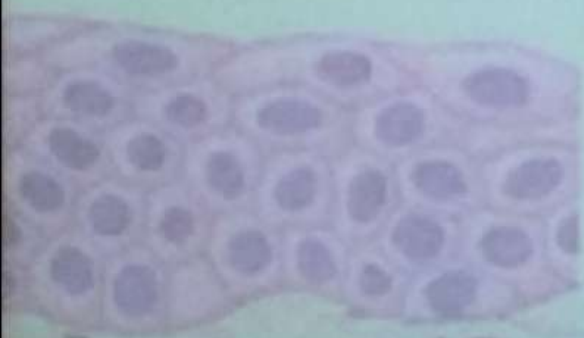
Relaxed bladder



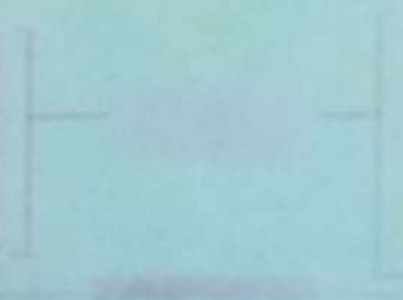
Relaxed bladder



LM x 400



Relaxed bladder



Relaxed bladder

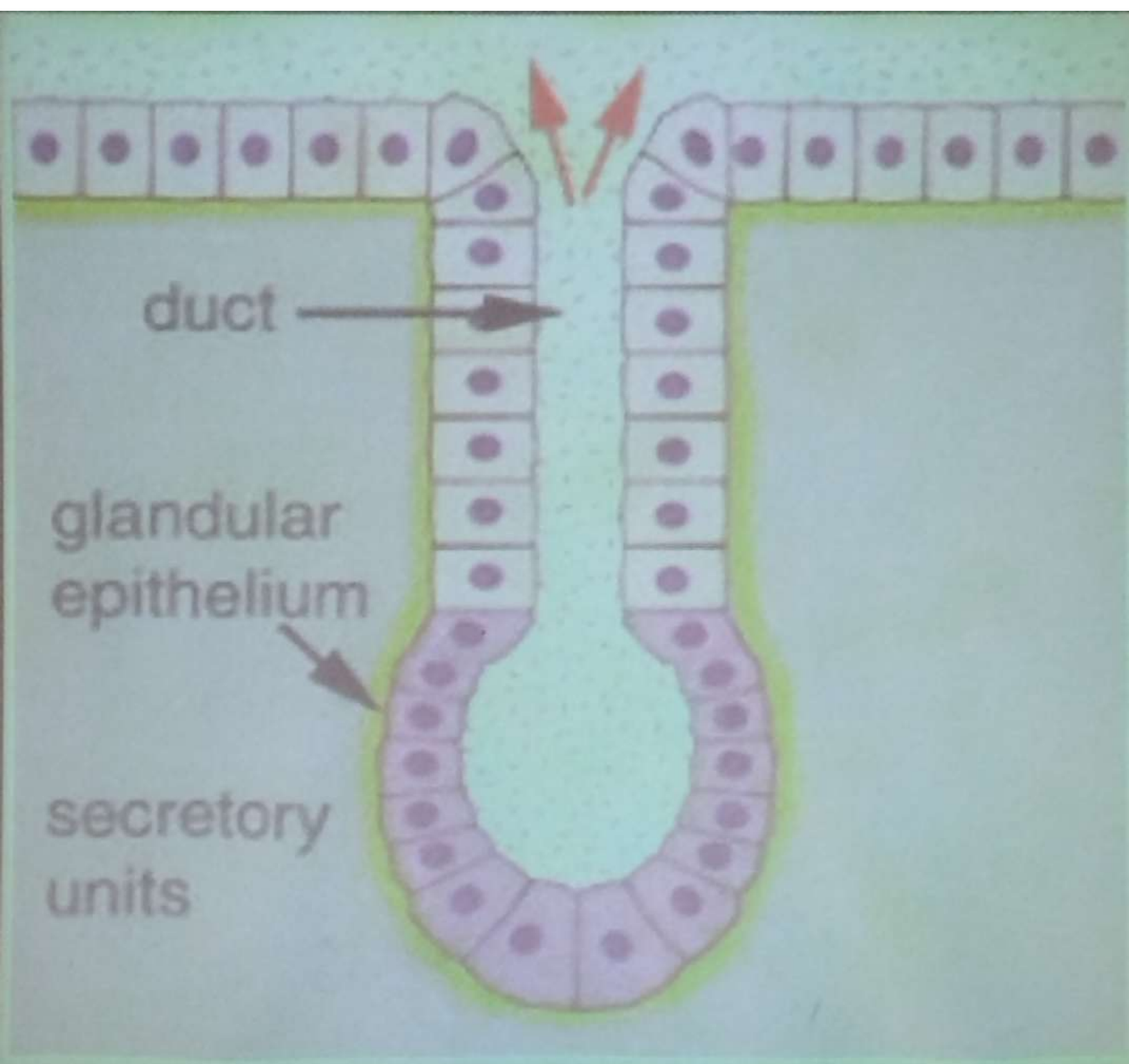


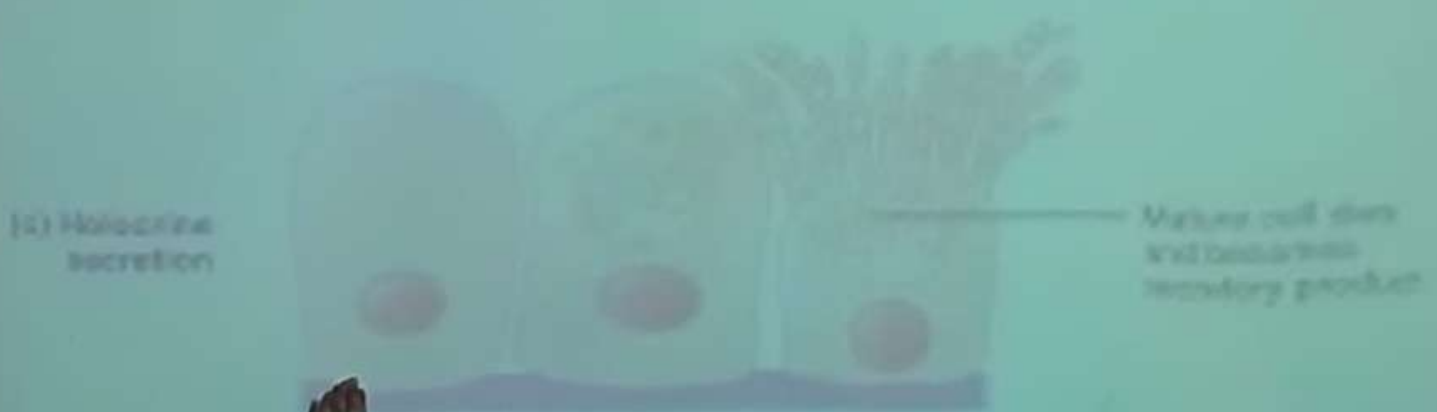
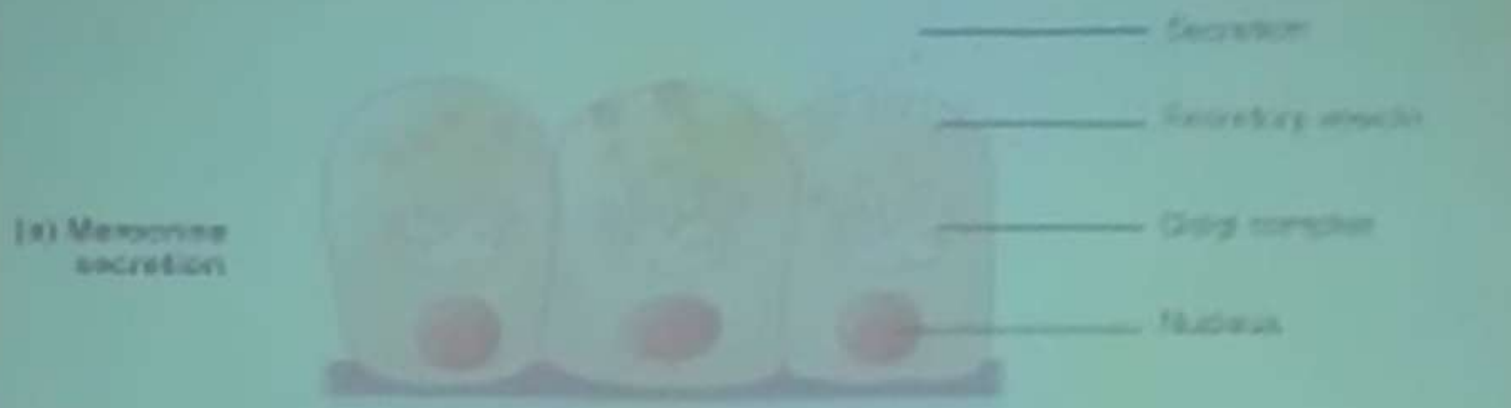
LM x 400

Glandular Epithelium

The glands of the body are classified as:

- 1. Endocrine glands.
- 2. Exocrine glands.
- 3. Mixocrine glands.





Classification according to nature of secretion

Serous gland:-

- The gland has small diameter, narrow lumen and deeply stain. The cells are pyramidal in shape with indistinct boundaries.. The nucleus are rounded and found near the center.
- -The cytoplasm is deeply basophilic at the base (due to localized ribosome (cytoplasmic RNA) and acidophilic zymogen granules are found near the free border .The myoepithelial cells (Basket cells) are less numerous than the mucous glands.